# Knowledge Distillation for Developing Versatile Controllers of Robotic Swarms

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**Abstract.** Designing robust, compact controllers capable of multi-task functionality remains a critical challenge for real-world swarm robotics systems (SRS). Traditional evolutionary robotics (ER) approaches often produce highly specialized controllers that lack the versatility needed for multiple tasks. This research proposes a knowledge distillation (KD) method for executing multiple types of tasks using the ER approach in SRS. We focus on a method that integrates multiple machine learning models trained on different tasks in image recognition using KD. Specifically, we conducted computer simulations where a swarm of mobile robots execute composite tasks by constructing a controller for each task (cooperative transport, obstacle avoidance, and aggregation), integrating these teacher models into a student model of the same size, and executing the composite task. Future work will focus on testing this method with real robots and larger swarms to see how it performs in real world environments. We also plan to explore how adding more sensors and improving real time learning can enhance the robots ability to handle more complex and unpredictable tasks.

**Keywords:** Swarm robotics · knowledge distillation · evolutionary robotics.

## 1 Introduction

Swarm robotic systems (SRS) are systems that utilize multiple relatively simple autonomous robots to generate emergent collective behavior from the interactions between individual robots [1, 2]. One method for constructing the control system in SRS is evolutionary robotics (ER) [3]. Most existing research that employs the ER approach in SRS targets simple environments and does not address environments with multiple types of tasks [4, 5].

To overcome this specialization gap and enhance controller versatility, this study proposes a novel transfer learning framework based on knowledge distillation (KD). In KD, a model with many layers and nodes, along with high prediction accuracy, is used as the teacher model, while a lightweight model to be embedded in the system serves as the student model. The teacher model's knowledge is then used for the student model's learning. The student model

is trained to minimize the soft target loss, which is the difference between the teacher model's output and the student model's output.

KD is a technique conventionally used in deep learning to transfer generalized knowledge from a large, complex teacher model to a smaller, more efficient student model [6]. In our unique adaptation, we prepare the same number of teacher models as the number of subtasks performed by the SRS, with each teacher model trained on one subtask. All the information from these teacher models is integrated into a single student model. Additionally, we aim to reduce performance disparities by adjusting the data volume according to the learning difficulty of each task.

The core contribution of this work are: (1) A principled KD framework for synthesizing a single, versatile swarm controller by aggregating heterogeneous expertise from multiple specialized ER controllers. (2) The introduction of an adaptive data adjustment mechanism that actively manages the training data volume from each teacher based on its initial learning accuracy. This mechanism is vital for mitigating the detrimental effects of inter task knowledge competition and disparate learning difficulty, thereby ensuring optimal integration of all required behaviors.

The remainder of this paper is structured as follows: Section 2 reviews related work. Section 3 details the design of the specialized teacher controllers and presents the proposed KD-based knowledge integration framework. Section 4.4 describes the experimental results, and Section 5 discusses the findings. Finally, Section 6 concludes the paper.

## 2 Related work

Evolutionary robotics remains a robust method for designing decentralized controllers [7, 8]. The ER paradigm, however, is fundamentally goal-specific; the evolved controllers are brittle when faced with tasks outside their training objective [9]. While approaches like modular robotics or hierarchical control offer solutions [10], they often increase architectural complexity or computational overhead, which is undesirable for simple swarm. Research on controller construction for multiple types of tasks often focuses on approaches using multi-objective optimization and model integration via transfer learning. In the approach utilizing multi-objective optimization proposed a method to find the optimal solution while considering the competition relationship between tasks [11].

On the other hand, in the approach using transfer learning proposed a method for computationally modeling the relationships between visual tasks and efficiently applying task transfer learning [12]. They analyzed the transfer dependencies between 26 types of 2D, 2.5D, 3D, and semantic tasks, and proposed a method for constructing a task classification map (taxonomy).This taxonomy clarifies the relatedness between tasks and demonstrated that transferring knowledge from related tasks can significantly reduce the required amount of labeled data while maintaining equivalent performance with about 2/3 data reduction compared to learning 10 tasks independently. This method uses a neural net-

work and explores the optimal combination for transfer learning without relying on conventional, intuitive assumptions about task relationships. Furthermore, it quantifies transferability between tasks and uses binary integer programming to calculate the optimal learning policy. In the research by Schaudt et al. [13], they constructed a model that estimates both the location and cause of pneumonia from chest X-ray (CXR) images by integrating teacher models each learning the location and cause of pneumonia separately into a student model using KD, a type of transfer learning. This research performed a three-stage KD process to classify different types of pneumonia (COVID-19, bacterial, fungal, other viral) and healthy cases. The details are as follows. They demonstrated that this method can improve the performance of pneumonia classification models even in limited data environments.Our work addresses the specialization challenge while maintaining a flat, homogeneous controller architecture.

Multi task learning (MTL) aims to improve performance across multiple tasks by sharing representations [14]. Recent work in reinforcement learning (RL) has explored policy distillation to transfer knowledge between agents. However, directly applying multi-objective optimization to ER often leads to trade-offs where one task dominates the others [15]. We frame the problem as knowledge fusion using KD, which allows us to decouple the high-performance training of specialized teachers from the final synthesis of the generalized student.

KD was introduced to compress large neural networks [6, 16]. In robotics, KD has been used for: (1) model compression, reducing policy size for deployment; and (2) Domain adaptation, transferring policies from simulation to reality [17]. Our innovation lies in using KD not for compression, but for knowledge aggregation from diverse, specialized sources into a single, compact, and multi-functional policy, allowing the student to simultaneously learn distinct control functions [18].

## 3 Knowledge integration method for SRS based on KD

In this research, as shown in Figure 1, we prepare the same number of teacher models corresponding to the number of subtasks performed by the SRS, with each teacher model trained on one subtask. All the information from these teacher models is integrated into a single student model. Additionally, we aim to reduce performance disparities by adjusting the data volume according to the learning difficulty of each task.

Determine the data increase ratio from the learning accuracy. If $A_{\max} = A_i$ is the accuracy of the model with the highest accuracy, $A_{i(\neq j)}$ is the accuracy of the $i$th teacher model $(i, j = 1, \ldots, N)$, $R_{\max} = R_j$ is the data increase ratio for the model with the highest accuracy, $R_{i(\neq j)}$ is the data increase ratio for the $i$th teacher model, and $P$ is the ratio coefficient, the increase ratio is expressed by the following equations:

$$\begin{cases} R_{\max} = 1 \\ R_i = P \times \frac{A_{\max}}{A_i} \end{cases} \tag{1}$$

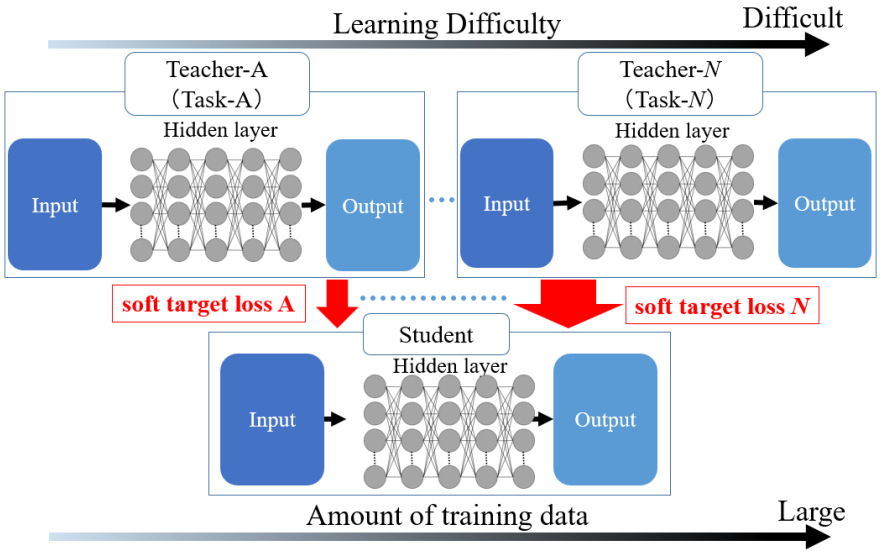
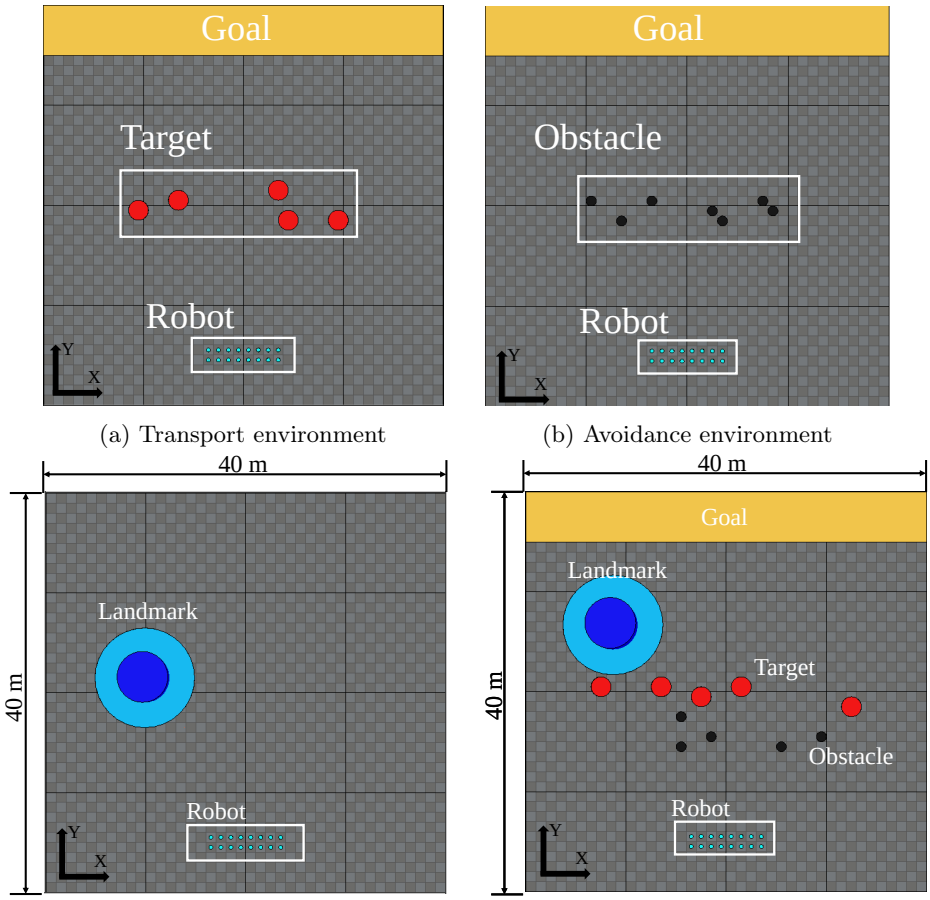



Fig. 1: Proposed method

The P-series models were developed using a novel data volume adjustment mechanism to selectively enhance learning for tasks where KD performance was initially weak. This mechanism utilizes $P$ to scale the amount of augmented training data. Model uses a ratio coefficient of $P = 0.5$, which represents a less aggressive. The $P = 1$ value sets the proportional baseline for the data increase, meaning the augmentation is directly proportional to the difference between the highest-performing teacher model and the specific teacher model's accuracy. Finally, the $P = 2$, which doubles the impact of the adjustment formula, resulting in a more aggressive increase in data volume. Comparing the performance of these different $P$ models helps determine the optimal balance for constructing a robust controller for the complex compound task.

## 4 Experiment settings

V-REP is adopted as the simulation environment for conducting the experiments in this research. V-REP is an open-source 3D robot simulation software. It is equipped with multiple physics engines such as ODE and Bullet. In this research, we use Bullet 2.78, which is the default setting of V-REP, as the physics engine and evolution strategies ($(\mu + \lambda)$-ES) algorithm with parameters ($\mu = 36, \lambda = 164$) is used.




(a) Transport environment (b) Avoidance environment

(c) Aggregation environment (d) Composite task environment

Fig. 2: Experimental environments for individual and composite task

### 4.1 Experimental environments

In this research, three types of tasks transportation, avoidance, and aggregation, as well as a composite task that includes all of them, are shown in Figure 2.

**(a) Transportation task** The experimental environment for the transportation task is shown in Figure 2(a). In the transportation task, the robot's objective is to transport the target to the goal. Let $n_{\mathrm{t}}$ be the number of targets (objects being transported), $D_{\mathrm{s}}$ be the initial distance between the target and the goal, and $D_{\mathrm{f}}$ be the final distance between the target and the goal. The fitness for the

transportation task, $F_{\mathrm{trs}}$, is expressed by Equation (2).

$$F_{\mathrm{trs}} = \sum^{n_{\mathrm{t}}} \frac{D_{\mathrm{s}} - D_{\mathrm{f}}}{D_{\mathrm{s}}} \times 100 \tag{2}$$

**(b) Avoidance task** The experimental environment for the avoidance task is shown in Figure 2(b). In the avoidance task, the robot's objective is to reach the goal, and points are subtracted based on the number of times the robot collides with an obstacle. Let $n_{\mathrm{o}}$ be the number of obstacles, $n'_{\mathrm{o}}$ be the number of robots that collided with an obstacle, $n_{\mathrm{r}}$ be the number of robots, $D'_{\mathrm{s}}$ be the number of robots that collided with an obstacle, and $D'_{\mathrm{f}}$ be the final distance between the robot and the goal. The parameter $w_{\mathrm{c}}$ represents the collision-penalty weight and controls how strongly collisions reduce the overall fitness. In the original formulation, this penalty weight is set to $w_{\mathrm{c}} = 3$, meaning each collision reduces the fitness by three points The fitness for the avoidance task, $F_{\mathrm{avo}}$, is expressed by Equation (3).

$$F_{\mathrm{avo}} = \sum^{n_{\mathrm{o}}} \sum^{n_{\mathrm{r}}} \frac{D'_{\mathrm{s}} - D'_{\mathrm{f}}}{D'_{\mathrm{s}}} \times 100 - n'_{\mathrm{o}} \times w_{\mathrm{c}} \tag{3}$$

**(c) Aggregation task** The experimental environment for the gathering task is shown in Figure 2(c). In the gathering task, the objective is for two or more robots to stay within the area (light blue range) around the landmark for a certain period of time. Let $n_{\mathrm{l}}$ be the number of robots that remained around the landmark, $t_i$ be the time each robot remained, and where $N_{\mathrm{min}}$ is the minimum robot count, $T_{\mathrm{max}}$ is the target time, $T_{\mathrm{min}}$ is the minimum time, and $S_{\mathrm{base}}$ is the base score. The fitness for the gathering task, $F_{\mathrm{agr}}$, is expressed by Equation (4).

$$F_{\mathrm{agr}} = \begin{cases} \frac{t_{\mathrm{i}}}{T_{\mathrm{max}}} \times S_{\mathrm{base}} \times n_{\mathrm{l}} & , n_{\mathrm{l}} \geq N_{\mathrm{min}} \wedge T_{\mathrm{min}} \leq t_{\mathrm{i}} < T_{\mathrm{max}} \\ S_{\mathrm{base}} \times n_{\mathrm{l}} & , n_{\mathrm{l}} \geq N_{\mathrm{min}} \wedge T_{\mathrm{max}} \leq t_{\mathrm{i}} \\ 0 & , \mathrm{otherwise} \end{cases} \tag{4}$$

The parameters $N_{\mathrm{min}}$, $T_{\mathrm{min}}$, $T_{\mathrm{max}}$, and $S_{\mathrm{base}}$ are set to 3, 5, 30, and 25, respectively, in this study.

**(d) Composite task** The experimental environment for the task is shown in Figure 2(d). The objective of the task is to transport the target to the goal while avoiding obstacles, and for two or more robots to stay around the landmark for a certain period of time. To clarify the fitness for each task, the evaluation is performed using Equation (2) for the transportation task, Equation (3) for the avoidance task, and Equation (4) for the gathering task.

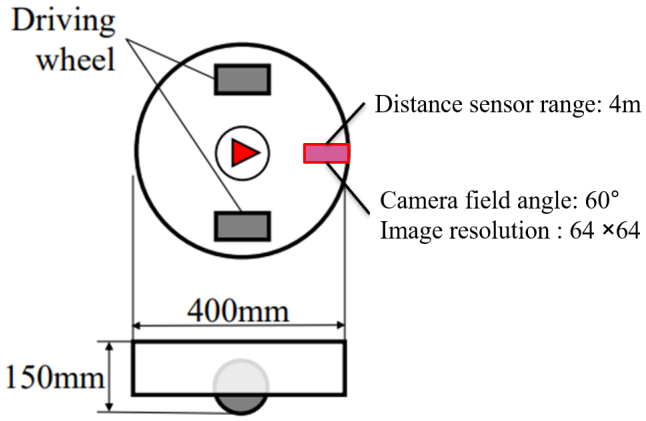



Fig. 3: Robot model

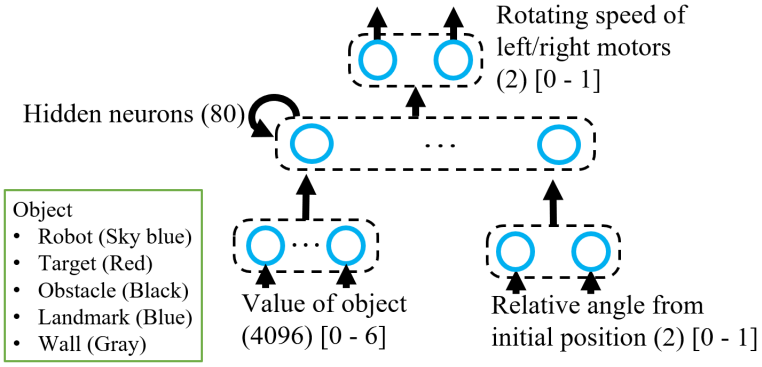



Fig. 4: Artificial neural network

### 4.2 Robot model

The robot is equipped with a camera and a distance sensor, both mounted on the front as shown in Figure 3. The camera has a range of 4 meters and provides an image with a resolution of 64 × 64 pixels. The field of view of the camera is 60°, and objects in the camera's view are identified based on their color. In addition to the camera, the robot also has a distance sensor with a detection range of 4 meters. The robot moves using a differential drive system with left and right driving wheels, allowing for precise control of its movement.

### 4.3 ANN configuration

The robot has a controller represented by an artificial neural network (ANN). Its structure is shown in Figure 4. The input layer of the ANN receives inputs

Table 1: Teacher model accuracy and increase rate.

| **Model** | **Accuracy[-]** | $\mathbf{P_{0.5}}$ | $\mathbf{P_1}$ | $\mathbf{P_2}$ |
|---|---|---|---|---|
| Transport-teacher | 0.195 | 1.35 | 2.70 | 5.40 |
| Avoidance-teacher | 0.527(= $A_{\max}$) | | 1 | |
| Aggregation-teacher | 0.230 | 1.15 | 2.30 | 4.60 |

Table 2: Accuracy of detected student models.

| **Model** | **Accuracy[-]** |
|---|---|
| Student | 0.807 |
| Adjustment | 0.827 |
| Adjustment(2) | 0.836 |
| Adjustment(0.5) | 0.812 |

from the camera sensor and 2 inputs representing the relative angle from the robot's initial position. The hidden layer has neurons and features a recurrent connection configuration, including self-connections for all nodes. The output layer outputs the target rotational speeds of the left and right motors.

### 4.4 Experimental results

The fitness of the models is compared and evaluated based on three types: the teacher model, the ER model, and the student model. The teacher model is trained for each specific task, while the ER model is created under the same experimental settings as the teacher model for the composite task. The student model integrates the teacher model using only KD, aiming to combine the benefits of the teacher models into a single, more versatile model.

Table 1 shows learning accuracy was highest for the avoidance teacher model, followed by the aggregation teacher model, and then the transportation teacher model. Table 2 shows higher learning accuracy compared to the teacher models. The results of the evaluation experiment in the composite task are shown in Table 3 and 4. The fitness for the transportation task was highest for the ER model, followed by the transportation teacher model, avoidance teacher model, adjustment model, aggregation teacher model, student model, 2× adjustment model, and 0.5× adjustment model, in descending order. The fitness for the avoidance task was highest for the 2× adjustment model, followed by the avoidance teacher model, ER model, aggregation teacher model, 0.5× adjustment model, student model, transportation teacher model, and adjustment model, in descending order. The fitness for the aggregation task was highest for the ER model, followed by the aggregation teacher model, transportation teacher model, 2× adjustment model, adjustment model, student model, 0.5× adjustment model, and avoidance teacher model.

Table 3: Average fitness (Teacher and ER model)

| **Task** | **Transport-teacher[-]** | **Avoidance-teacher[-]** | **Aggregation-teacher[-]** | **ER[-]** |
|---|---|---|---|---|
| Tra. | 37.7 | 34.6 | 32.2 | 46.3 |
| Avo. | 196 | 356 | 349 | 354 |
| Agg. | 8.89 | 3.69 | 11.3 | 13.9 |
| Total | 243 | 394 | 393 | 414 |

Table 4: Average fitness (Student and adjustment model)

| **Task** | **Student[-]** | **Adjustment[-]** | **Adjustment(2)[-]** | **Adjustment(0.5)[-]** |
|---|---|---|---|---|
| Tra. | 26.9 | 33.2 | 24.6 | 15.2 |
| Avo. | 313 | 146 | 525 | 331 |
| Agg. | 5.93 | 6.37 | 8.73 | 4.07 |
| Total | 346 | 186 | 558 | 350 |

## 5 Discussion

We summarize the results of comparing the average fitness for each task across three trials for each model. Here, $F_{\text{tra-ave}}$ is the average fitness for the transportation task, $F_{\text{avo-ave}}$ is the average fitness for the avoidance task, and $F_{\text{agg-ave}}$ is the average fitness for the aggregation task.

The ER model shows a high tendency in the average fitness of all tasks. Additionally, the corresponding teacher model (e.g., transportation teacher model for the transportation task) also shows a high tendency, suggesting that learning through ER remains effective even when the number of task types is multiple. When comparing the task-specific teacher models with the student model, the teacher model showed higher fitness than the student model in all tasks. This is likely because the interference from information regarding other tasks, introduced during model integration, hindered the learning of the student model.

The fitness of each model is presented as a box plot, and the statistical significance determined by the t-test is shown in Figure 5. The data increase ratio for the transportation task in the $P_{0.5}$ model is 1.35; we believe this was insufficient, leading to a bias in the created data. This likely impeded learning and caused a decline in performance in the transportation task.

Since the $P_1$ model increased the data volume by 2.70 times, we believe this increased the number of learning iterations, effectively improving the performance of the transportation task. Conversely, a significant difference was observed in the student model for the avoidance task. This is likely because, while the data volume for transportation and aggregation increased, the data for the avoidance task remained unchanged, meaning the avoidance data relatively decreased as a proportion of the total data, leading to a reduction in learning volume for avoidance. Next, we compare the student model and the $P_2$ model. Although the $P_2$ model had the highest data increase ratio for the transportation and aggregation tasks, no significant difference was observed in the $P_2$ model for

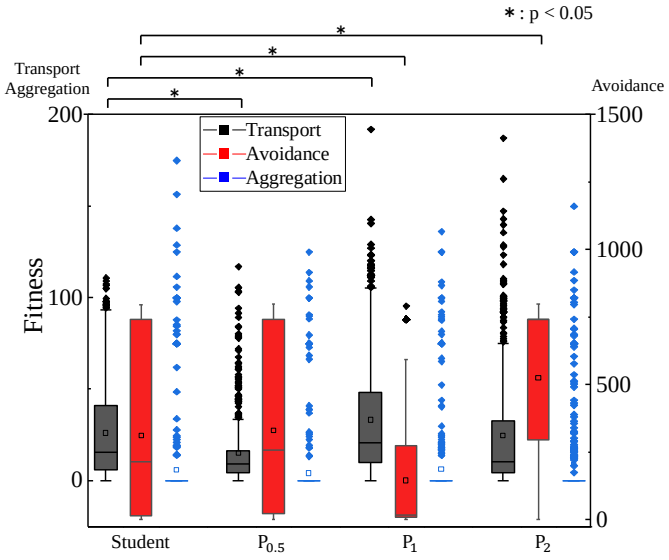
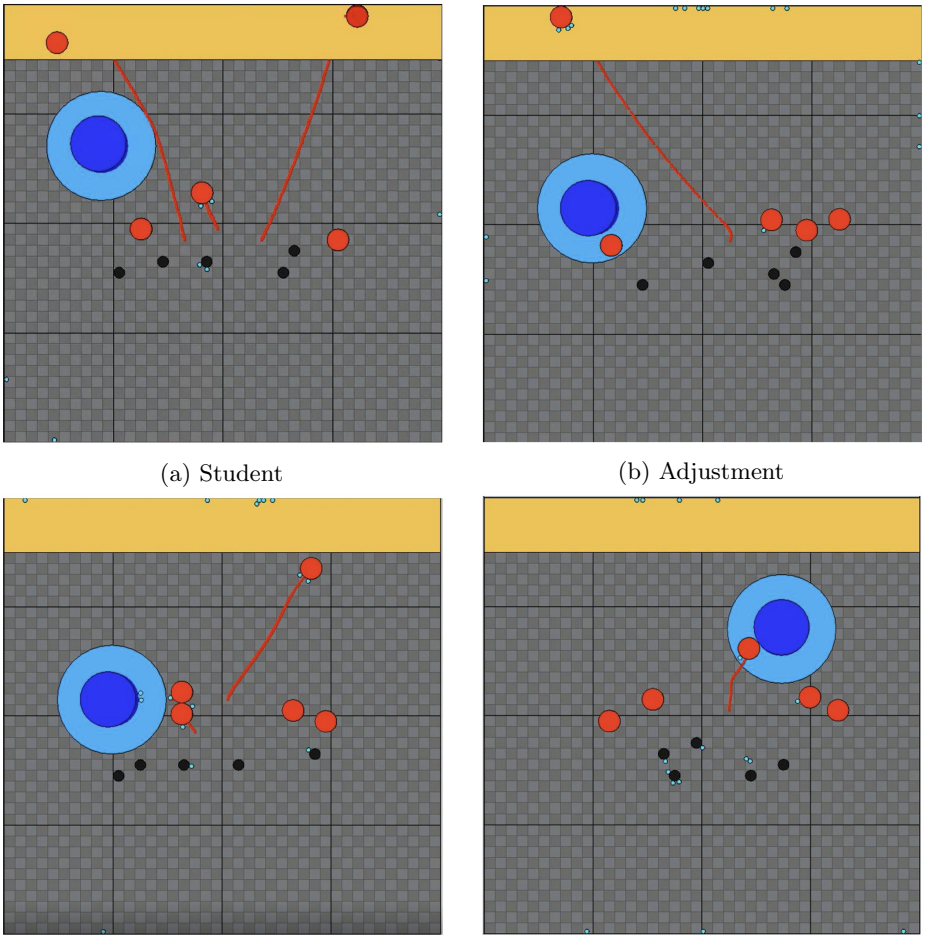



Fig. 5: Significantly different by t test

these tasks. This is thought to be because the data for transportation and aggregation tasks accounted for over 90% of the total training data, but the learning gradients of the transportation and aggregation teacher models differed, leading to competition in learning that prevented effective knowledge transfer. However, a significant difference was observed in the $P_2$ model for the avoidance task. Table 5 shows the behavior ratios of the teacher model and the $P_2$ model in the composite task. Among the teachers model behaviors, forward movement is the most frequent shown in Figure 6. In the $P_2$ model, forward movement is also the most frequent, and such behavior is less likely to cause competition, making it easier to acquire during learning. Furthermore, forward movement showed the highest ratio in the avoidance teacher model, suggesting that forward movement is effective in increasing the avoidance task fitness.

Table 5: Behavior ratio of teacher and $P_2$ model [%]

| **Model** | **All teachers** | **Avoidance-teacher** | **$P_2$** |
|---|---|---|---|
| Forward | 43.9 | 65.1 | 99.8 |
| Right turn | 20.0 | 33.7 | 0.08 |
| Left turn | 5.19 | 0.03 | 0.06 |
| Right rotate | 12.9 | 0.08 | 0.02 |
| Left rotate | 2.09 | 1.06 | 0.00 |
| Stop | 15.9 | 0.00 | 0.00 |



(a) Student (b) Adjustment

(c) Adjustment(2) (d) Adjustment(0.5)

Fig. 6: A series of adjustments to refine and finalize the setup.

## 6 Conclusion

In this research, to execute multiple types of tasks in a swarm robotics system (SRS), we developed a controller for each task and integrated their knowledge into a student model of the same size as the teacher models using knowledge distillation (KD), thereby creating a model capable of performing composite tasks. Furthermore, considering that the learning accuracy varies for each task during integration, we developed a mechanism to adjust the training data volume of each teacher model.

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